

1.) Steady-state ; 1-D conduction

a.) $k_{Al} = 237 \text{ W/mK}$

$L = 500 \mu\text{m} = 5 \times 10^{-4} \text{ m}$

Fouriers law for single layer $T_{s1} = 305$; $T_{s2} = 295$

$$q'' = k \frac{\Delta T}{L} = 237 \frac{\text{W}}{\text{mK}} \left(\frac{10 \text{ K}}{5 \times 10^{-4} \text{ m}} \right) = 4.74 \times 10^6 \frac{\text{W}}{\text{m}^2}$$

$L = 500 \text{ nm} = 5 \times 10^{-7} \text{ m}$

$$q'' = k \frac{\Delta T}{L} = 237 \left(\frac{10}{5 \times 10^{-7} \text{ m}} \right) = 4.74 \times 10^9 \frac{\text{W}}{\text{m}^2}$$

b.) $k_{\text{air}} = .026 \text{ W/mK}$

$L = 500 \mu\text{m} = 5 \times 10^{-4} \text{ m}$

$$q'' = k \frac{\Delta T}{L} = .0263 \frac{\text{W}}{\text{mK}} \left(\frac{10 \text{ K}}{5 \times 10^{-4} \text{ m}} \right) = 5.2 \times 10^2 \frac{\text{W}}{\text{m}^2}$$

$L = 500 \text{ nm} = 5 \times 10^{-7} \text{ m}$

$$q'' = k \frac{\Delta T}{L} = .0263 \left(\frac{10 \text{ K}}{5 \times 10^{-7}} \right) = 5.2 \times 10^5 \frac{\text{W}}{\text{m}^2}$$

c.) Thermal Resistance E_2

$$R_{\text{tot}} = \frac{L_s}{k_w} + \frac{L_{\text{air}}}{k_{\text{air}}} + \frac{L_s}{k_{\text{Al}}}$$

$$q'' = \frac{\Delta T}{R_{\text{tot}}}$$

$L_s = 40 \mu\text{m}$

$L_{\text{air}} = 420 \mu\text{m}$

$$R_{\text{tot}} = \frac{40 \times 10^{-6}}{237} + \frac{420 \times 10^{-6}}{.0263} + \frac{40 \times 10^{-6}}{237} = 1.615 \times 10^{-2} \frac{\text{m}^2 \text{K}}{\text{W}}$$

$$q'' = \frac{10 \text{ K}}{1.615 \times 10^{-2}} = 6.19 \times 10^2 \frac{\text{W}}{\text{m}^2}$$

$$R_{\text{tot}} = \frac{40 \times 10^{-7}}{237} + \frac{420 \times 10^{-7}}{.0263} + \frac{40 \times 10^{-7}}{237} =$$

$L_s = 40 \text{ nm}$

$L_{\text{air}} = 420 \text{ nm}$

$$= 1.615 \times 10^{-5}$$

$$q'' = \frac{10}{1.615 \times 10^{-5}} = 6.19 \times 10^5 \frac{\text{W}}{\text{m}^2}$$

2.) Given: $T_{\infty} = 25^{\circ}\text{C}$
 $h = 25 \text{ W/m}^2\text{K}$
 $\rho_i = 920 \text{ kg/m}^3$
 $L_f = 334 \text{ kJ/kg} = 3.34 \times 10^5 \text{ J/kg}$
 $k_w = 0.5 \text{ W/mK}$
 $H(\text{Ice thickness}) = 0.01 \text{ m}$
 $(\text{film thickness}) = x(t)$

a.)

$$\rho_i L_f \frac{dx}{dt} = q''(t) = \frac{T_{\infty} - 0}{\frac{1}{h} + \frac{x(t)}{k_w}}$$

$$\frac{dx}{dt} = \frac{T_{\infty}}{\rho_i L_f \left(\frac{1}{h} + \frac{x}{k_w} \right)} \Rightarrow \int_0^s \left(\frac{1}{h} + \frac{x}{k_w} \right) ds = \frac{T_{\infty}}{\rho_i L_f} t$$

$$\frac{s}{h} + \frac{s^2}{2k_w} = \frac{T_{\infty}}{\rho_i L_f} t$$

$$x(t) = k_w \left[-\frac{1}{h} + \sqrt{\left(\frac{1}{h} \right)^2 + \frac{2T_{\infty}}{\rho_i L_f k_w} t} \right]$$

b.)

$$t_m = \frac{\rho_i L_f}{T_{\infty}} \left(\frac{H}{h} + \frac{H^2}{2k_w} \right)$$

$$\frac{\rho_i L_f}{T_{\infty}} = \frac{(920)(3.34 \times 10^5)}{25} = 1.229 \times 10^7 \text{ J/m}^2\text{K}$$

$$t_m = 1.71 \text{ hrs}$$